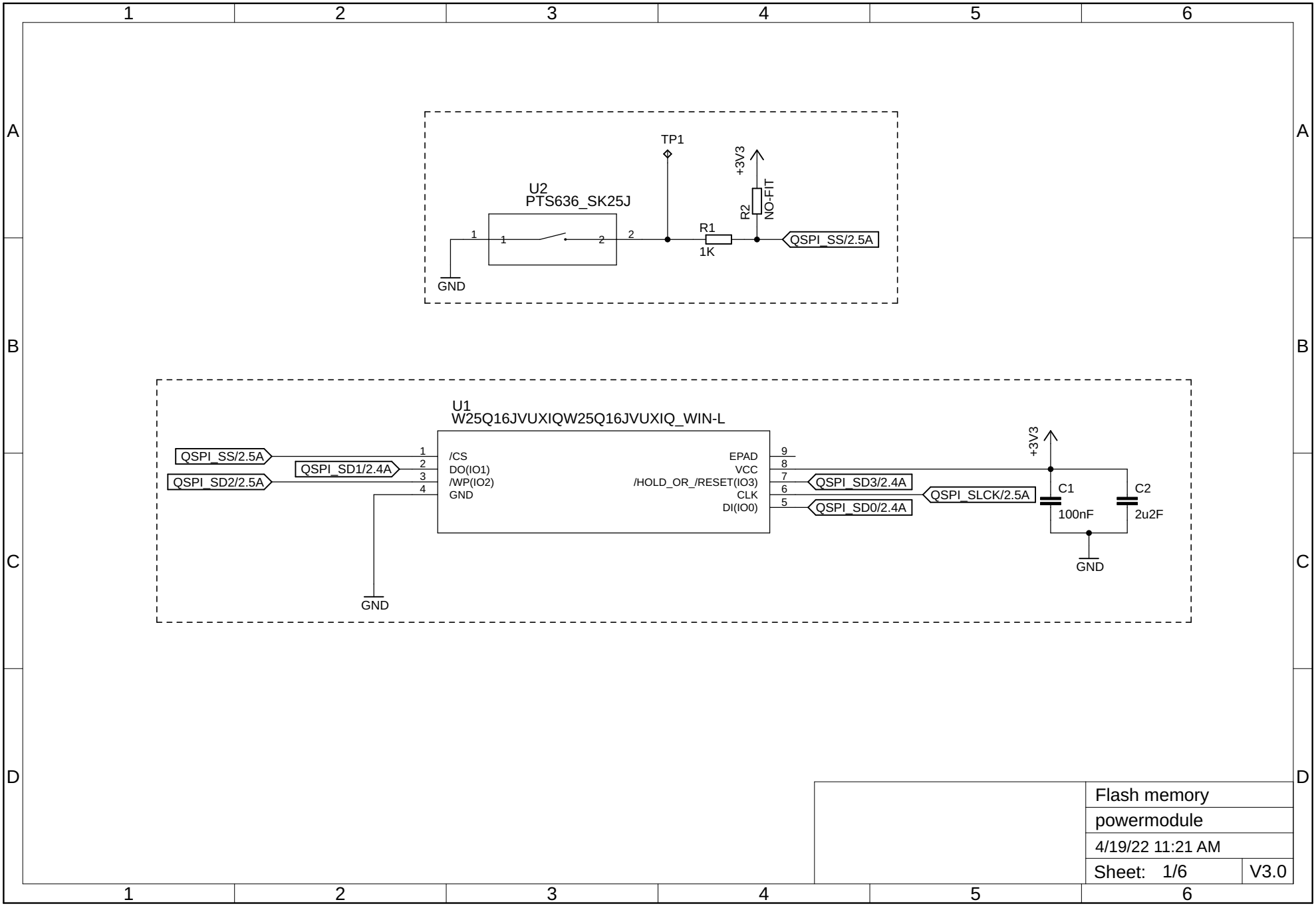
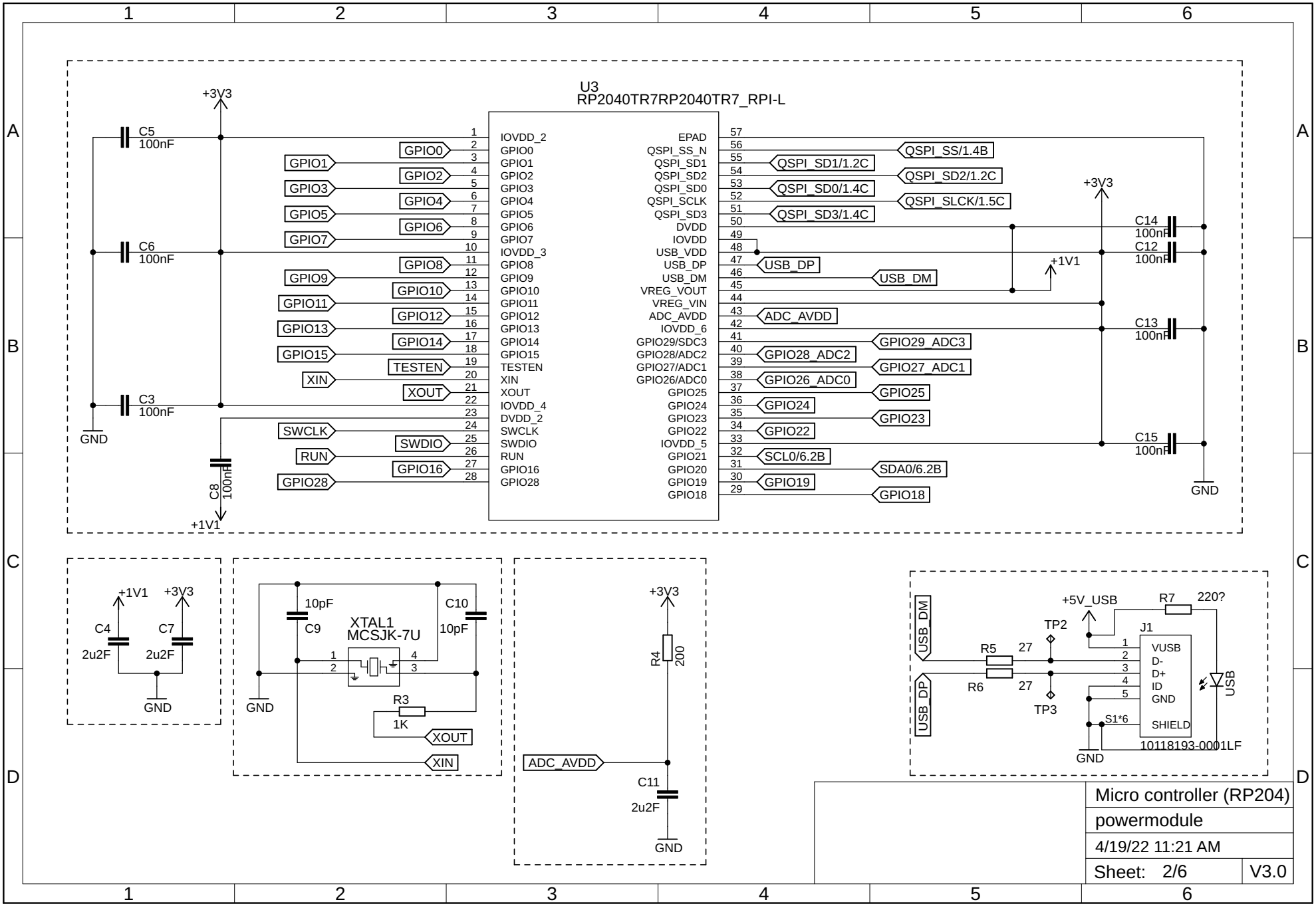
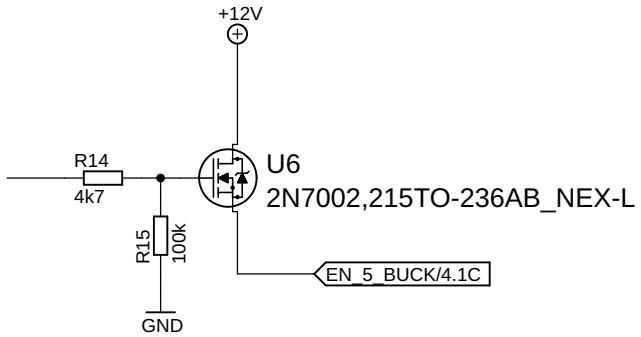
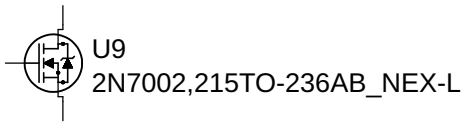
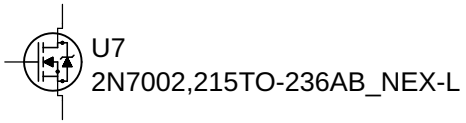
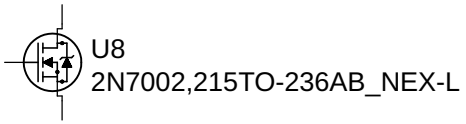

1
2
3
4
5
6
A
B
C
D
TP1
+3V3
U2
PTS636_SK25J
R2
NO-FIT
R1
1K
QSPI_SS/2.5A
GND
U1
W25Q16JVUXIQW25Q16JVUXIQ_WIN-L
QSPI_SS/2.5A
QSPI_SD1/2.4A
QSPI_SD2/2.5A
/CS
DO(IO1)
/WP(IO2)
GND
EPAD
VCC
/HOLD_OR_/RESET(IO3)
CLK
DI(IO0)
QSPI_SD3/2.4A
QSPI_SLCK/2.5A
QSPI_SD0/2.4A
C1
100nF
C2
2u2F
Flash memory
powermodule
4/19/22 11:21 AM
Sheet: 1/6
V3.0


U3
RP2040TR7RP2040TR7_RPI-L
+3V3
C5
100nF
C6
100nF
C3
100nF
GND
C8
100nF
+1V1
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
TESTEN
XIN
XOUT
SWCLK
SWDIO
RUN
GPIO16
GPIO28
IOVDD_2
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
IOVDD_3
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
TESTEN
XIN
XOUT
IOVDD_4
DVDD_2
SWCLK
SWDIO
RUN
GPIO16
GPIO28
EPAD
QSPI_SS_N
QSPI_SD1
QSPI_SD2
QSPI_SD0
QSPI_SCLK
QSPI_SD3
DVDD
IOVDD
USB_VDD
USB_DP
USB_DM
VREG_VOUT
VREG_VIN
ADC_AVDD
IOVDD_6
GPIO29/SDC3
GPIO28/ADC2
GPIO27/ADC1
GPIO26/ADC0
GPIO25
GPIO24
GPIO23
GPIO22
IOVDD_5
GPIO21
GPIO20
GPIO19
GPIO18
QSPI_SS/1.4B
QSPI_SD1/1.2C
QSPI_SD2/1.2C
QSPI_SD0/1.4C
QSPI_SLCK/1.5C
QSPI_SD3/1.4C
USB_DP
USB_DM
ADC_AVDD
GPIO29_ADC3
GPIO28_ADC2
GPIO27_ADC1
GPIO26_ADC0
GPIO25
GPIO24
GPIO23
GPIO22
SCL0/6.2B
SDA0/6.2B
GPIO19
GPIO18
+3V3
+1V1
C14
100nF
C12
100nF
C13
100nF
C15
100nF
GND
+1V1
+3V3
C4
2u2F
C7
2u2F
GND
10pF
C9
XTAL1
MCSJK-7U
C10
10pF
GND
R3
1K
XOUT
XIN
+3V3
R4
200
ADC_AVDD
C11
2u2F
GND
USB_DM
USB_DP
+5V_USB
R7
220?
TP2
R5
27
R6
27
TP3
J1
VUSB
D-
D+
ID
GND
SHIELD
USB
S1*6
10118193-0001LF
GND
Micro controller (RP204)
powermodule
4/19/22 11:21 AM
Sheet: 2/6
V3.0

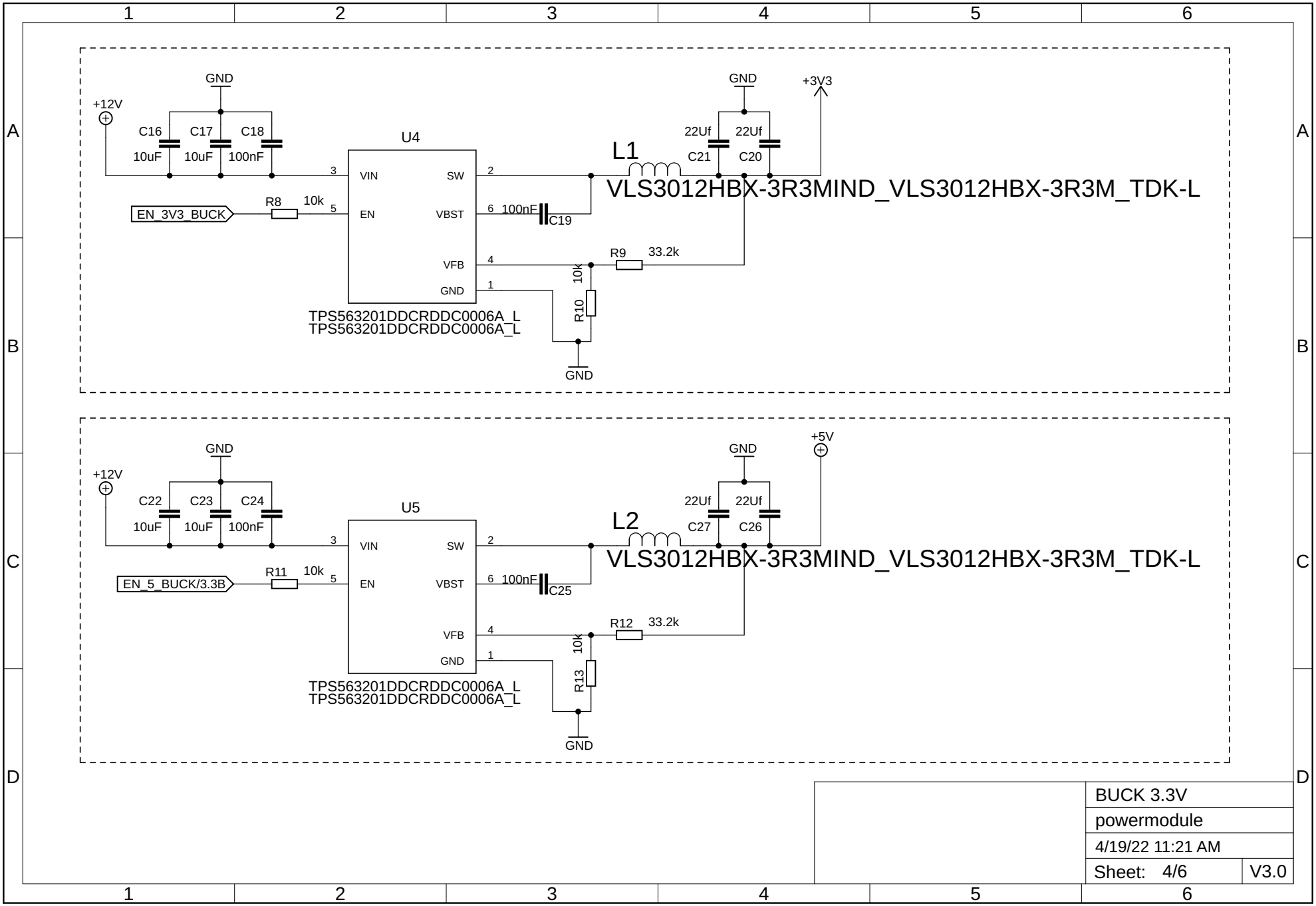


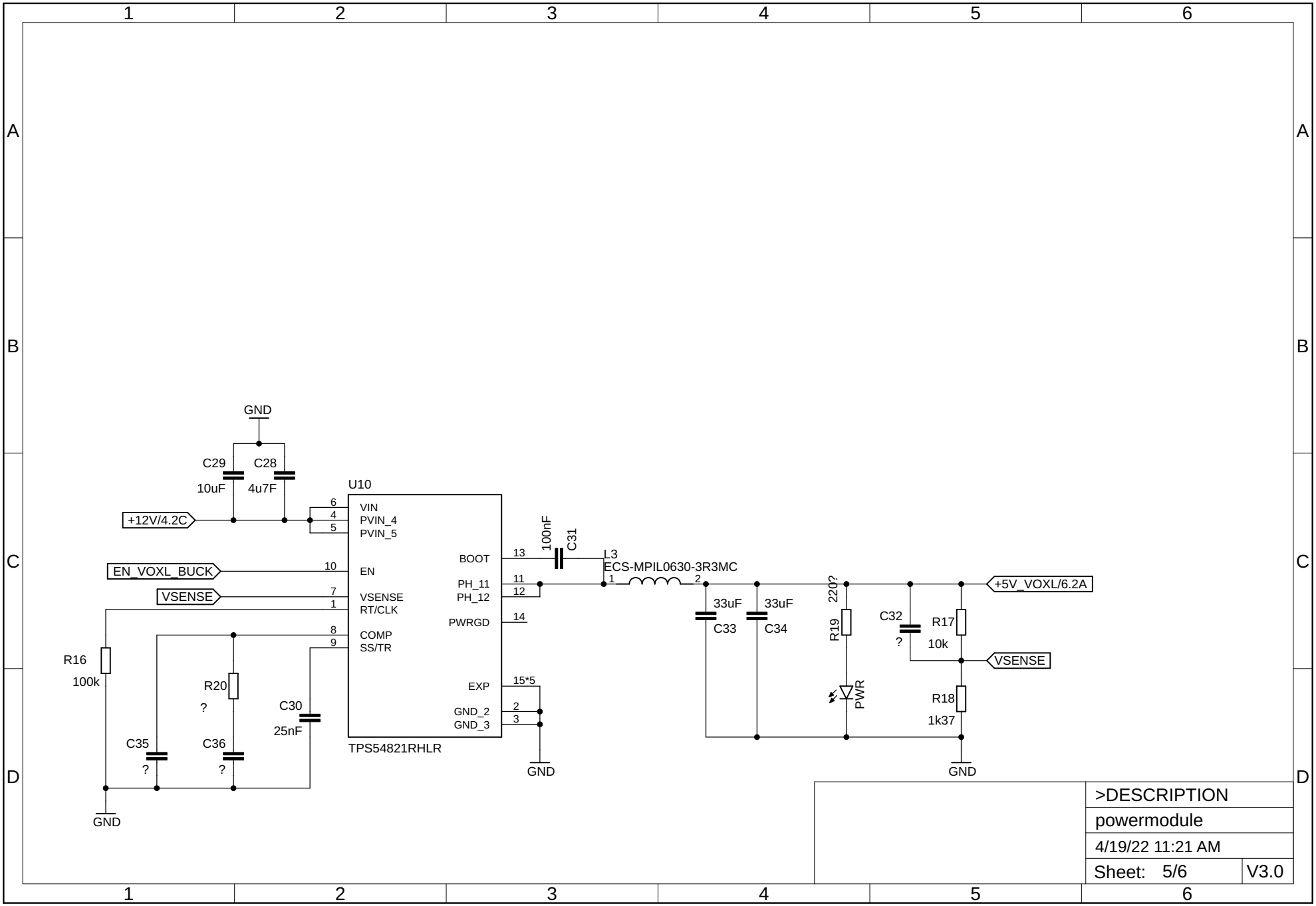


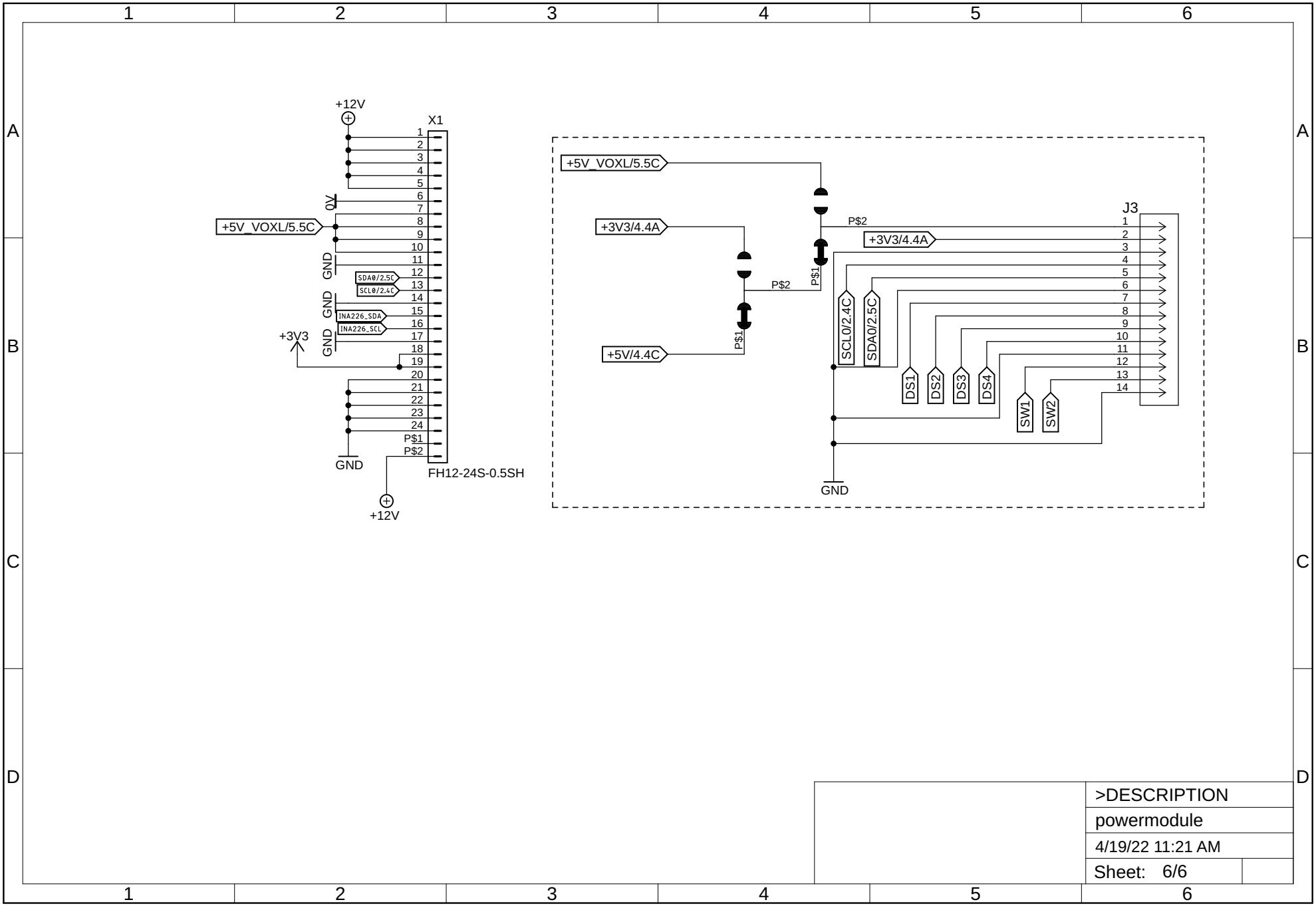




>DESCRIPTION	
powermodule	
4/19/22 11:21 AM	
Sheet: 3/6	V3.0


GND
+12V
C16
C17
C18
10uF
10uF
100nF
U4
VIN
SW
EN
VBST
VFB
GND
EN_3V3_BUCK
R8
10k
100nF
C19
L1
VLS3012HBX-3R3MIND_VLS3012HBX-3R3M_TDK-L
22Uf
22Uf
C21
C20
+3V3
R9
33.2k
10k
R10
TPS563201DDCRDDC0006A_L
TPS563201DDCRDDC0006A_L
GND
+5V
C22
C23
C24
U5
EN_5_BUCK/3.3B
R11
C25
L2
C27
C26
R12
R13
BUCK 3.3V
powermodule
4/19/22 11:21 AM
Sheet: 4/6
V3.0


GND
C29
10uF
C28
4u7F
U10
+12V/4.2C
6
4
5
VIN
PVIN_4
PVIN_5
100nF
C31
13
BOOT
L3
ECS-MPIL0630-3R3MC
EN_VOXL_BUCK
10
EN
11
PH_11
12
PH_12
1
2
33uF
C33
33uF
C34
220?
R19
+5V_VOXL/6.2A
VSENSE
7
1
VSENSE
RT/CLK
14
PWRGD
C32
?
R17
10k
VSENSE
R16
100k
8
9
COMP
SS/TR
R20
?
C30
25nF
EXP
15*5
2
3
GND_2
GND_3
PWR
R18
1k37
C35
?
C36
?
TPS54821RHLR
GND
GND
GND
>DESCRIPTION
powermodule
4/19/22 11:21 AM
Sheet: 5/6
V3.0


+12V
X1
0V
+5V_VOXL/5.5C
GND
SDA0/2.5C
SCL0/2.4C
GND
INA226_SDA
INA226_SCL
+3V3
GND
GND
P$1
P$2
FH12-24S-0.5SH
+12V
+5V_VOXL/5.5C
+3V3/4.4A
+5V/4.4C
P$2
P$1
P$2
P$1
J3
+3V3/4.4A
SCL0/2.4C
SDA0/2.5C
DS1
DS2
DS3
DS4
SW1
SW2
GND
>DESCRIPTION
powermodule
4/19/22 11:21 AM
Sheet: 6/6